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Optimization of an air-cooled heat sink for cooling of a solar photovoltaic panel: A computational study

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ABSTRACT

Rapidly increasing global energy demand and resulting climate change effects have enhanced the integration of renewable systems such as solar photovoltaics into the built environment. For countries with hot climates, despite vast applications and numerous technological advancements, lower efficiencies of the solar panels due to high temperature is still a major disadvantage affecting the feasibility of using photovoltaic technology. This study analyses the cooling performance of air-cooled heat sinks for the climate of Dubai, UAE, using Computation Fluid Dynamics (CFD). A stepwise optimization study was conducted to investigate the effects of varying fin spacing, baseplate thickness, fin height and fin thickness on the heat dissipation rate. Further, the elemental properties of the heat sink were studied by altering its material to aluminium. The base-model heat sink could reduce PV cell temperature by 27 °C in an ambient temperature of 42 °C. The optimized fin spacing, baseplate thickness, fin height and fin thickness of 7, 0.0025 m, 0.12 m and 0.002 m further reduced the average panel temperatures by 3.5%, 4% and 9% respectively. The use of copper as an effective heat sink material was concluded compared to aluminium that conversely increased the average panel temperature by 2% despite the use of optimized heat sink structure.

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1. Introduction

The issues of energy crisis and global warming have become increasing alarming with the socio-economic development and growing population. Buildings account for a major percentage of the global energy consumption and carbon emissions through construction and operation. The 2004 IPCC report stated that buildings are responsible for 30% of the anthropogenic carbon emission worldwide [22]. Many efforts have been made to decrease the building energy consumption through various energy-efficient measures and adopting renewable energy generation strategies. Therefore, more and more buildings are focused on reducing the operational energy consumption by the integration of renewable energy generation that uses clean energy from natural sources that are constantly replenished [30].Table 1.

Renewable energy generation has grown over the past decade with hydro, wind and solar being the most reliable and widely used renewable power sources [14]. Solar power being the most abundant resource proves to be the third largest renewable energy source for electricity generation, and heating purposes in buildings. The zero greenhouse gas emissions and global resource endowment properties have made solar energy the best alternative to meet the world's escalating energy demand [31]. Energy from solar power is mainly drawn by Photovoltaic Technology that converts sunlight directly into electricity through photovoltaic effect of silicon cells, a semiconducting material [27]. The International Environmental Agency (IEA) stated that PV technology could supply about 25% of the global electricity by 2050 [10]. As a result, the installed PV capacity in top developed countries doubles every year [19]. Meanwhile, the IRENA foresees the building related CO_2 emissions to be offset by around 32 Gigatons per year by 2050 through the integration of renewables [33].

Rise in PV cell temperature has two major consequences: rapid decline in generated electrical performance (output power and efficiency) and thermal fatigue that significantly reduces lifespan of PV systems. Besides, the silicon cells itself produce heat as all other electronics thereby, intensifying the issue [10]. Non-uniform operating temperatures and heat spots also cause loss of power and irreversible thermal degradation [21]. Despite various technological advancements, only 20% of the incident solar radiation hitting the cell surface gets converted to electricity and more than 50% of it gets dissipated as excess heat [32,33]. According to experimental results, the relative electrical efficiency of a PV modules reduces by ~0.4–0.5% for every 1° rise in temperature, as shown by the graph below [36,23]. High ambient temperatures

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Table 1
Physical configuration of base model copper heat sink.

Physical Component	Specification
Fin Height	0.03 *m*
Distance between fins	0.05 *m*
Number of fins	9
Length = Height of heat sink	0.50 *m*
Baseplate Thickness	0.002 *m*
Fin Thickness	0.002 *m*
Fin Length	0.50 *m*

cause the negatively charged stray currents to flow through the embedded parts of the panel (glass, mount & frame) causing corrosion of the silicon cells and permanent damage [37]. The lead-acid batteries also proved to be disadvantageous in hot climates as higher electrochemical activities within the battery foster corrosion, diminishing its capacity and total lifespan [13]. As a result, it is extremely crucial to reduce the operating temperature of photovoltaic cells to achieve maximum optimum yield and protect the cells from irreversible damage.

To improve the power generation performance and maintain total lifespan of solar panels various active and passive cooling techniques have been continuously studied and developed. A successful cooling system must significantly increase the module's efficiency in a much lower payback period. The cooling systems are mostly classified into active and passive. Active cooling systems consume additional power during the heat dissipation process by facilitating cooling via specialized flow ducts or spraying water on panel surfaces [20,25]. Whereas passive cooling techniques draw out heat without any additional power consumption, through natural conduction or convection. The medium in both the systems can be air, water (natural coolant), or artificial refrigerants [15,2]. The two cooling strategies are competing methods for heat dissipation; however, the latter is found to be more feasible in application as the additional power consumption and added costs in active cooling minimize the benefits of net gain in efficiency from cooling [6].

The use of air-cooled heat sinks with protruding fins is a common passive cooling method that facilitates heat transfer through natural convection. It acts as a heat exchanger that carries away heat from PV surface to the surrounding medium (air) through its fins that are specifically designed to maximize the area for heat transfer [17,16,18]. Heat sinks are characterized based on rate of heat removal and temperature distribution whereas its cooling performance is essentially governed by physical configurations, elemental properties of the metal, and ambient temperatures [38,33,23].

This study numerically analyses the cooling performance of the most optimum heat sink structure for a standalone photovoltaic panel using the ANSYS Fluent CFD software. An optimization study is carried out on the physical configuration of this base model to enhance the heat dissipation rate of the heat sink. Effectiveness of the models is depicted by examining the temperature reduction at the surface and surrounding domain of the PV module. Lastly, the efficiency of copper as a preferred heat sink material is evaluated by comparing it against aluminium. Effectiveness of the resulting heat sink structure is directly measured through theoretical calculations of overall efficiency improvement in the passively cooled monocrystalline PV panel.

Popovici et al. [26] carried out a numerical study towards the passive cooling of a photovoltaic panel using an air-cooled heat sink to attain the cooling effect by altering the angle between ribs and base plate. A PV panel with rectangular-finned copper heat sink was modelled into CFD ANSYS software to study the cooling effect under a solar radiation of 500 W/m^2 and an external temperature of 35 °C. Numerical simulation for this turbulent model (Re = 13100) estimated the intensity of heat transfer and concluded that the cooling performance of heat sinks increased with increase in ribs height but decreased with increase in their angle of inclination. The angles studied were 45°, 90° and 135°, at three different heights of 0.01 m, 0.03 m and 0.05 m. The PV cell temperature dropped by 10 °C on average but, the heat sink with 0.05 m rib height and 45° angle of inclination, depicted maximum temperature drop of 16 °C, from 56 °C to 39.9 °C. This numerical approach presented an energy-efficient solution for cooling the PV panels as a 7.55% and 6.97% rise in power production was observed for 45° and 90° rib angle, respectively.

To maximize this cooling potential of metal heat sinks Cui et al., [7] conducted a numerical parametric study on the physical structure of rectangular finned air-cooled heat sink for a single concentrator PV cell using Icepak CFD software. Starting with fin spacing, comparing 6 fin numbers (from 2 to 12) with fin height and thickness of 1.4 cm, and 1.5 mm, depicted optimal number of fins to be 8, as the lowest operating temperature of 41 °C was obtained, at an ambient temperature of 30 °C. Study of the effect of four different base thicknesses of the heat sink on heat dissipation concluded that the temperatures of CPV system only slightly changed with the increase in base thickness. However, an extremely large thickness was stated undesirable as it causes material wastage and obstructs heat rejection. Likewise, the analysis of varying fin thicknesses also did not show vast differences in operating temperatures of the CPV system. Effects of varying fin heights were studied to observe that the temperatures of the solar cell decreased with increasing fin height, due to increased area for heat rejection. The average solar cell temperature with optimum heat sink structure was 36.5 °C. These optimal dimensions proved to be extremely useful in developing heat sinks for standalone concentrator solar cells.

Arifin et al. [4] experimentally validated the application of an air-cooled heat sink on a PV module simulation that showed a reduction of up to 10 °C in the average PV temperature when irradiated with 1000 W/m^2 solar energy at ambient temperature of 35 °C. The experimental setup of aluminium heat sink with perforated fins of 2 mm thickness also verified the improvements in measured electrical efficiency and output power by 2.6% and 18.67%.

Based on the ongoing research on heat sink application for photovoltaic panels it is found that metallic (copper or aluminium) and rectangular finned air-cooled heat sinks facilitate effective cooling and improve the electrical performance of PV panels, however the physical structure and surrounding microclimate are important factors affecting its optimum cooling performance [12].

2. Aim and objectives

The main aim of this research is to develop and investigate a heat sink structure with optimal parameters that facilitates the cooling of a PV panel under the microclimate of Dubai, thereby maximizing its efficiency and power output.

The objective of this study is to develop the most optimal heat sink structure in terms of its fin spacing, baseplate thickness fin height, fin thickness and material using ANSYS-Fluent CFD software. Additionally, to investigate the air temperature and velocity at the surface and surrounding domain of the passively cooled PV panel and to determine the overall efficiency and power output performance of a passively cooled monocrystalline PV panel using the optimum heat sink structure.

Although, there is extensive existing research on the passive cooling of PV panels using metal heat sinks, it is limited to lower ambient temperatures of around 30–35 °C, unlike the climate of

Dubai with high year-round average temperature of 42 °C and annual solar radiation of 2285 kWh/m^2 [5] Further, there is minimal research on enhancing the structural and elemental properties of the heat sinks for cooling standalone PV panels. Therefore, this paper performs a numerical parametric study using computational fluid dynamics on a previously validated heat sink model by Popovici et al. [26] which results in the most efficient and high performing heat sink model.

3. Methodology

To achieve the objectives of developing an effective heat sink, it is necessary to apply modern approaches to the design which requires analysis of existing models and optimization of the existing heat exchange properties. It is important to adopt a time and cost-effective method. While carrying out experiments is too expensive and often limited due to lack of equipment, CFD simulations provide virtual modelling tools and high quality solutions for evaluation of heat exchangers without actual time, expenses and physical conditions required [34]. The CFD simulations provide highly accurate result of the nature of fluid flow and heat distribution around the heat sink which is necessary to calculate the heat sink performance [8,29].

This section primarily describes the numerical analysis of a heat sink equipped PV module to observe its cooling performance on a typical summer day in Dubai. A parametric study on a validated heat sink base model was then performed to maximize its heat transfer rate and cooling performance by varying the physical configurations in the order of fin spacing, baseplate thickness, fin height and lastly fin thickness [7]. Each set of parametric study concludes an optimum geometrical configuration that is used as a benchmark model for the subsequent study. Lastly, the most optimum heat sink structure was further examined by altering its material to aluminium to conclude an effective metal that will be advantageous in fully utilizing the cooling properties of the proposed heat sink. The influence of optimum air-cooled heat sink structure on the electrical performance of a PV module was directly measured by calculating increases of efficiencies and higher power output under the same conditions of solar radiation. Figs. 1 and 2.

3.1. Validated heat sink model

To validate the simulation results of this numerical study, the heat dissipation rate of the base model heat sink was compared to a similar numerical study by Popovici et al., [26]. The researched model of a copper heat sink from exisitng literature was replicated

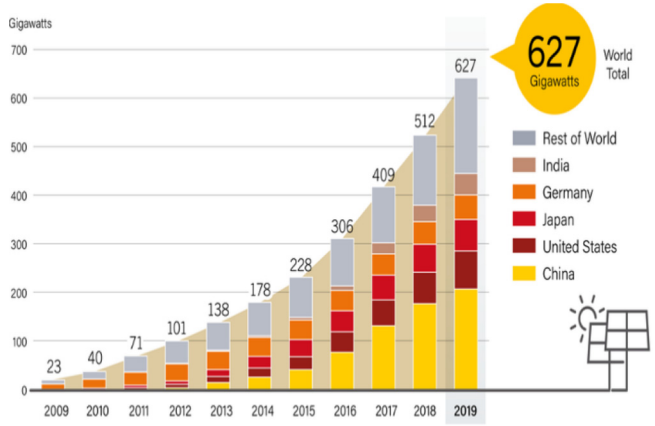


Fig. 1. Global capacity of installed PV technology, [19].

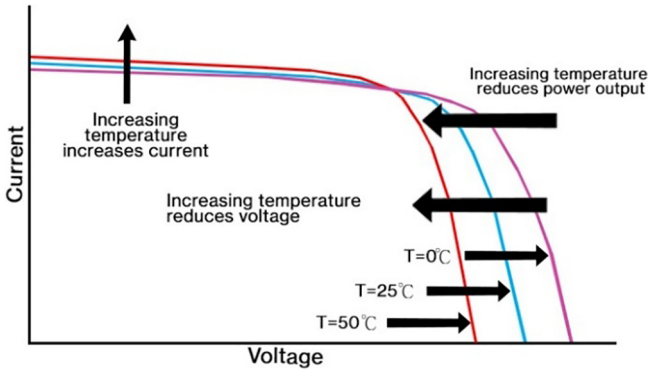


Fig. 2. Effect of high operating temperatures on PV panel performance, [20,35].

into ANSYS Fluent database to compare its cooling efficiency under lower ambient temperature and normal solar radiance of 35 °C and 500 W/m^2 respectively. The heat sink model depicted a maximum temperature drop of 22 °C, from 57 °C to 35 °C, and cooler operating temperatures of 43 °C were obtained, as seen in Fig. 3 below. The lower cooling performance for validation model, compared to that of base model (27 °C) was a result of a much cooler ambient environment only and thus validated the accuracy of simulation results. Likewise, the high turbulence region at the center of the airflow channel illustrated the maximum heat transfer zone close to the fin geometry. This region however has a slightly lower velocity of 2.53 m/s compared to the base model (2.55 m/s). Hence, the simulation results of base model heat sink were found to have acceptable accuracy.

3.2. Numerical analysis

This numerical study was carried out using mesh-based ANSYS Fluent CFD software using Navier-Stokes equations for simulating turbulent models of heat sinks. The mesh-based solver divides the model into simpler zones allowing in-depth analysis of numerical models for local approximation of larger domains to obtain precise simulation results [10]. Whereas the following governing equations of mass, momentum and energy conservation applied to cell zones to further add accuracy to the turbulent models with irregular fluid motion [1].

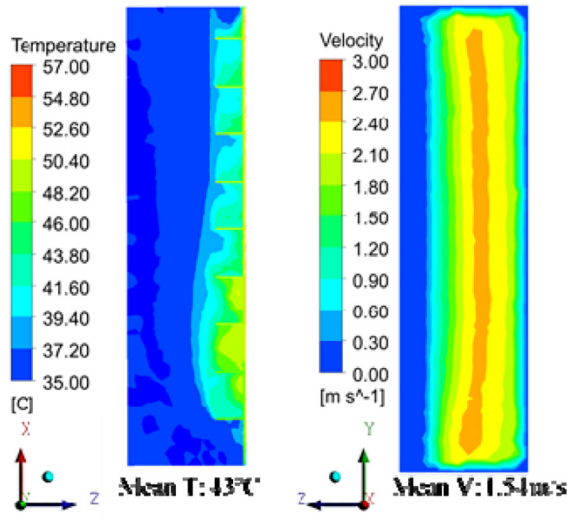


Fig. 3. Temperature (left) and velocity (right) contours of validated heat sink model by Popovici et al., [26].

Continuity Equation:

$$\nabla \vec{V} = 0 \tag{1}$$

Momentum Equation:

$$\frac{\partial \vec{V}}{\partial t} + \left(\vec{V} \cdot \nabla\right)\partial \vec{V}_{=\frac{1}{\rho}} \nabla_p + \upsilon \nabla^2 \vec{V} + p\vec{f} \tag{2}$$

Energy Equation:

$$pC_p\left[\frac{\partial T}{\partial t} + \left(\vec{V}.\nabla\right)T\right] = k\nabla^2 T + \varphi \tag{3}$$

Where,

$\vec{V}$ = velocity vector (m/s), p = specific mass (kg/m^3), ϑ = kinematic viscosity of fluid (m^2/s), $\vec{f}$ = force per unit mass (N/kg), T = temperature (K), K = thermal conductivity (W/m^{2}K), ϑ = viscous dissipation.

The model also used Solar Ray Tracing tool from Fluent software to simulate solar radiation incident on the PV panel. This ray tracing algorithm applied to the external PV cell layer uses sun position vectors and illumination parameters to compute resulting heat fluxes particularly on the boundary zones [3]. The average solar intensity prevalent in Dubai was manually added to this tool which acts as a uniformly spread heat source over the PV panel model.

The simulation of base model heat sink followed by the configuration and material parametric studies along with the geometry, computational mesh and boundary conditions input are specifically explained in sections 3.3, 3.4 and 3.5.

3.3. Base model heat sink

Geometry:

The geometry of simulation model was created in SolidEdge (Version 2021) and exported to ANSYS DesignModeler. The physical domain consisted of a vertical free-standing photovoltaic panel and a copper heat sink, both surrounded by an enclosure of fluid medium (air) as seen in Fig. 4. The 0.1 m wide air channel was created to investigate the nature of airflow and convectional heat transfer close to the geometry of fins. The dimensions of the PV panel being studied were: L(length) = H (height) = 0.5 m. The panel geometry was simplified as a single PV cell layer instead of multiple constituent layers as it created inaccurate conformal zones in Fluent solver. The copper heat sink was modelled with the following dimensions derived from existing numerical research conducted by Popovici et al. [26].

Computational mesh:

The mesh generation of physical domain was realized with automatically generated unstructured triangular mesh with fine refinement of 5°. This yielded a high-quality mesh with maximum skewness of 0.91, 62,234 nodes and 248,985 elements. A skewness level below 0.95 is recommended for triangular mesh with 60° cell shape which will otherwise destabilize the solution causing convergence difficulties in fluent solver. The large value of nodes and elements in the meshed model depicts smaller cell size and hence detailed local approximation of the geometry that corresponds to better accuracy in simulation results. Also, use of unstructured mesh type reduced the risk of oversimplified geometry resulting in a lesser efficient mesh distribution and a high cell count [1]. The mesh generated for heat sink integrated PV model is shown in Fig. 5 and the detailed mesh quality description for each parametric study is depicted in Table 2.

Grid Verification:

Verification of the numerical model was performed through grid adaptation to check the stability of the computational operation. The validation was carried out by using five different mesh configurations ranging between 5,072 and 135,455 elements. This was followed by a comparison of the computed average microclimate velocity and temperature in all the cases. The velocity and temperatures were measured along the outlet plane as indicated in Fig. 4 above.

The variation of the measured variables and their associated values of posteriori error are depicted in Fig. 6. The measured variables are consistent in magnitude as shown in Fig. 6a. The error values drop during the last two stages of grid adaptation as indicated in Fig. 6b, indicating that the CFD model is fairly stable across all the stages of grid adaptation.

Boundary conditions:

The 3D steady-state analysis of the model was initiated by enabling energy equation. The k-ε Renormalization Group (RNG) model of turbulence was used particularly for the air flow inside the ventilated channel to demonstrate superior capturing of fluid flow than standard k-ε turbulence model [3]. The model used a direct solar radiance of 900 W/m^2, with a disadvantageous ambient temperature of 42 °C under the Solar Ray Tracing tool.

Also, to apply the effect of high temperatures, the semitransparent layer of PV cells is synthesized with a global absorption coefficient of solar radiation, α = 0.6. The corresponding Reynold's number calculated for this imposed velocity was 13100, using the Eq. (4). The turbulent intensity of 4.8% was then derived using the Eq. (5). The recommended hydraulic diameter of 0.166 m is additionally assigned to support the calculations of turbulent airflow through the ventilated channel [26].

$$R_e = \rho_{\frac{VL}{\mu}} \tag{4}$$

$$I = 0.16.R_e^{-\frac{1}{8}} \tag{5}$$

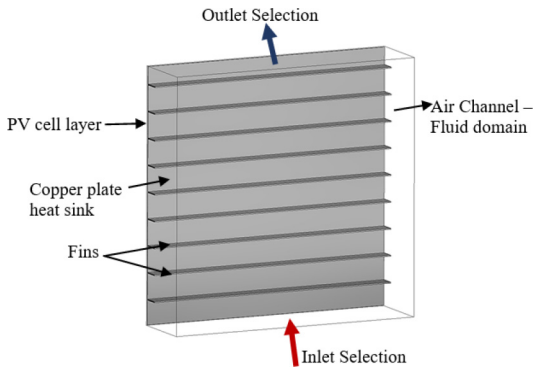


Fig. 4. Physical geometry of simulated model.

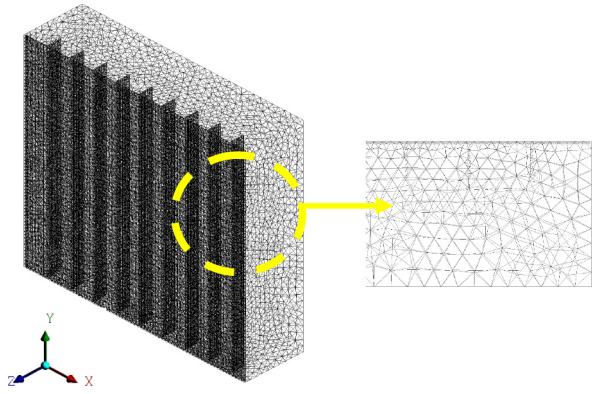


Fig. 5. Computational mesh configuration of simulated model – Unstructured triangular mesh.

Table 2
Mesh quality description for each parametric study.

Case	Physical component (m)	Nodes	Elements	Maximum Skewness
Case 1- Varying fin spacing	a) 0.03	71,787	276,380	0.90
	b) 0.04	63,421	251,082	0.90
	c) 0.05	62,234	248,985	0.91
	d) 0.06	58,508	234,476	0.91
	e) 0.07	55,673	221,842	0.91
	f) 0.08	51,566	204,903	0.91
Case 2- Varying baseplate thickness	a) 0.002	58,508	234,476	0.93
	b) 0.0025	58,285	233,952	0.93
	c) 0.003	57,754	231,825	0.93
	d) 0.0035	56,551	226,439	0.93
	e) 0.004	56,678	227,155	0.93
Case 3- Varying fin height	a) 0.03	58,285	233,952	0.93
	b) 0.06	58,508	234,476	0.93
	c) 0.09	89,339	371,449	0.93
	d) 0.12	96,378	397,084	0.93
Case 4- Varying fin thickness	a) 0.002	96,378	397,084	0.93
	b) 0.0025	96,459	394,059	0.93
	c) 0.003	98,723	397,243	0.93
	d) 0.0035	71,795	392,321	0.93
	e) 0.004	71,925	392,901	0.93

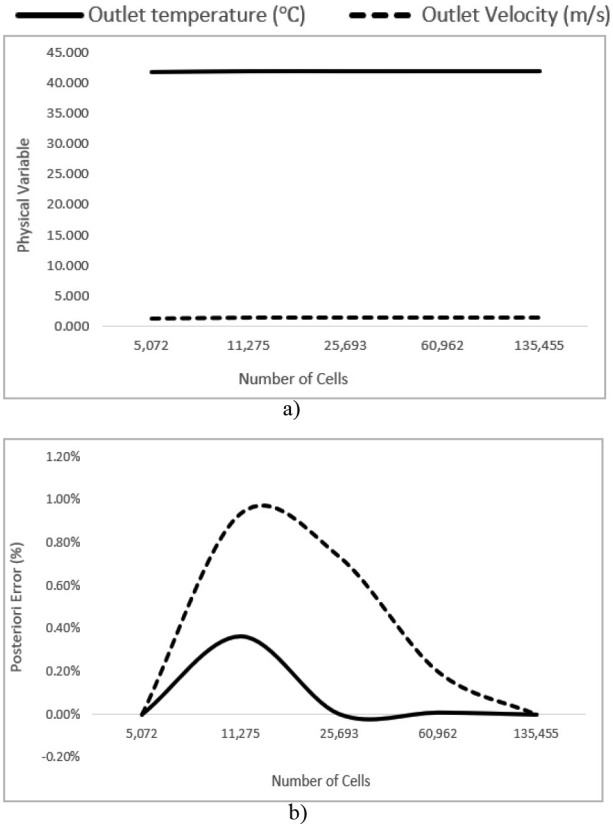


Fig. 6. (a) Variation in the measured variables with cell count; (b) Variation in the measured posteriori error with cell count.

The applied boundary conditions for all geometrical faces are further synthesized in Tables 3 and 4 and material inputs for their respective geometries are specified in Table 5.

3.4. Parametric studies on physical configuration

3.4.1. Case 1 – Varying fin spacing

The spacing between each fin was varied from 0.03 m to 0.08 m keeping the fin height, fin thickness, and baseplate thickness constant (at 0.03 m, 0.02 m, 0.02 m respectively). A total of 6 models were produced in the same way as the preliminary model to study the effect of fin spacing and number of fins on the heat dissipation rate and air convection.

Solution methods:

Double-precision version of Fluent computational tool was used to solve the governing equations of continuity, momentum, and energy. Since the geometry involves intersection of fluid enclosure with two layers (PV cell and heat sink), use of double-precision solver is necessary to resolve small pressure differences between the layers that derive the flow. The fluid flow and heat transfer problems were solved using finite volume method and SIMPLE pressure-velocity coupling that involved First Order Upwind to ensure better accuracy. This method uses the coupling relationship between velocity and pressure to implement mass conservation and derive the pressure fields [9]. The solution yielded a convergence criterion of 10^{-6} for the energy equations and 10^{-3} for pressure, velocity, and continuity equations.

Geometry:

The geometry was also similarly developed in SolidEdge and exported to ANSYS Designmodeler. Fig. 7 shows the geometries of PV panel equipped heat sink for 6 different fin spacings:

Table 3
Applied boundary conditions in ANSYS Fluent.

Description	Turbulent intensity (%)	Hydraulic diameter (m)	Velocity (m/s)	Backflow temperature (°C)	Solar transmissivity factor
Inlet	4.8	0.166	1.5	42	1
Outlet	4.8	0.166	–	42	1

Table 4
Cell boundary conditions applied to walls.

Description	BC Type	Convection	Radiation	Solar transmissivity factor
External PV cell layer	Semi-transparent	8 W/m^{2}k	Absorptivity factor = 0.6	1
Walls	Opaque	–	N/A	1
Walls – Fluid domain	Semi-transparent	–	N/A	1

Table 5
Material input in ANSYS Fluent.

Material	Geometry	Thickness (m)	Thermal conductivity (W/m.K)	Density (kg/m^3)	Specific heat capacity (J/kg.K)
Copper	Heat Sink	0.02	387.60	8978	381
PV cells	PV Panel	225×10^{-6}	148	2330	677
Air	Airflow channel	0.10	0.02	1.22	1006.43

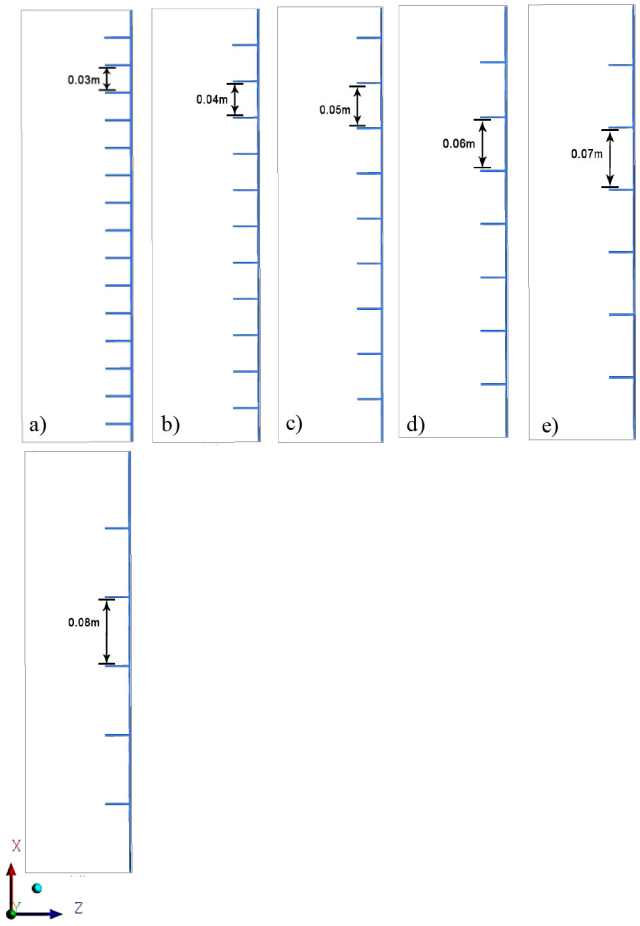


Fig. 7. Geometry of simulated heat sink integrated PV models with varying fin spacing a)0.03 m, b)0.04 m,c) 0.05 m, d)0.06 m e)0.07 m and f)0.08 *m.*

0.03 m, 0.04 m, 0.05 m, 0.06 m, 0.07 m and 0.08 m respectively. Fig. 8.

3.4.2. Case 2 – varying baseplate thickness

The most advantageous performance configuration from above models was applied to the next analysis of optimum baseplate thickness by analyzing 5 different thicknesses – 0.002 m, 0.0025 m, 0.003 m, 0.0035 m and 0.004 m keeping the fin spacing, fin height and fin thickness constant at 0.06 m, 0.03 m and 0.002 m. The models were similarly produced using the below mentioned geometry and mesh specifications.

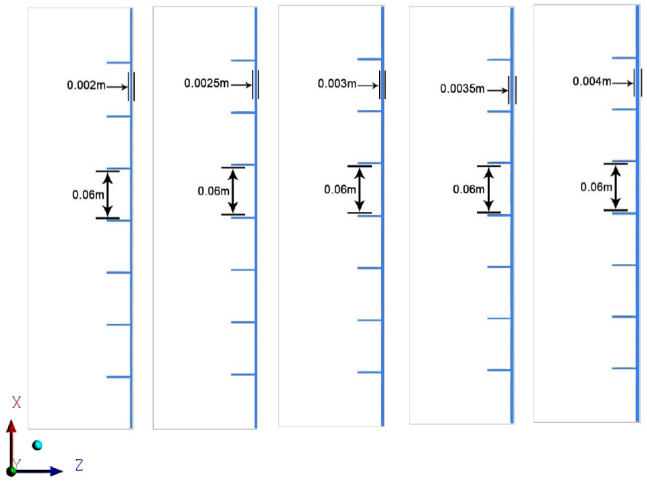


Fig. 8. Geometry of simulated heat sink integrated PV models with varying base thickness a)0.002 m, b)0.0025 m,c) 0.003 m, d)0.0035 m and e)0.004 *m.*

Table 6
Elemental properties of aluminium used for CFD simulation.

Elemental Properties	Parameters
Thermal conductivity (W/m.K)	202.4
Density (kg/m^3)	2719
Specific heat capacity (J/kg.K)	871

Geometry:

The ANSYS Designmodeler geometries used for this parametric study are as follows.

3.4.3. Case 3 – Varying fin height

The optimum baseplate thickness was applied to the configuration study of most advantageous fin height. Cooling performances of these four models with increasing fin heights from 0.03 m to 0.12 m were studied.

Geometry:

The geometrical configuration inputs of the models in ANSYS Designmodeler are as follows. The height was increased by 1% for each simulation, keeping the fin arrangement, base thickness, and fin thickness constant at 0.06 m, 0.0025 m and 0.002 m respectively.

3.4.4. Case 4 – Varying fin thickness

Lastly, the fin thickness of 0.002 m was increased by 0.25% to obtain the most optimum fin geometry with high structural stability and efficient cooling performance.

Table 7
X and Z axes coordinates of the lengths.

	Axial	Tangential
X- coordinates	Point 1- (0); Point 2- (0.50)	Point 1- (0.25); Point 2- (0.25)
Y-coordinates	Point 1- (0.005); Point 2- (0.005)	Point 1- (0.25); Point 2- (0.25)
Z- coordinates	Point 1- (-0.015); Point 2- (-0.015)	Point 1- (-0.01); Point 2- (-0.1)

Geometry:

Geometrical configuration inputs of the models in ANSYS Designmodeler are as follows. The fin thicknesses of 0.002 m, 0.0025 m, 0.003 m, 0.0035 m and 0.004 m were investigated for this parametric study.

3.4.5. Case 5 – Material properties

The elemental properties of the most optimum heat sink structure were further investigated using CFD analysis by altering its material to aluminium with the elemental properties in Table 6. The material input of final heat sink model was changed, keeping the geometry, mesh specifications, and boundary conditions constant as explained in the above sections.Table 7.

3.5. Determination of electrical performance of PV panel

The influence of each optimized heat sink parameter was directly measured by theoretical calculations of efficiency and power output for a polycrystalline PV panel having nominal efficiency (STC) of 16.20% at a cell temperature of 25 °C [11]. Calculations were performed based on the fact that the electrical efficiency of a PV module reduced by 0.5% for every 1 °C rise in temperature [36]. The power output was subsequently calculated using Eq. (6) [28], wherein the daily incoming solar radiation of 900 W/m^2 and panel area of 0.25 m^2 were used. The differences in electrical performance of the optimized models and base model thus validated the advantageous optimized parameters by clearly explaining the enhanced cooling performances.

$$Power = Radiation \times Area \times efficiency \quad (6)$$

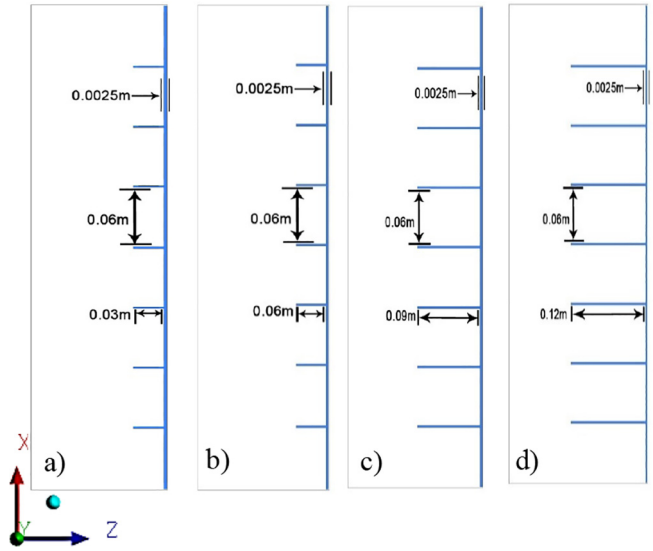


Fig. 9. Geometry of simulated heat sink integrated PV models with varying fin height a)0.03 m, b)0.06 m,c) 0.09 m and d)0.12 *m.*

4. Results

4.1. Base model heat sink

Simulation results from ANSYS Post-processor consisted of temperature and velocity contours representing heat transfer rate and airflow from the PV panel surface to the surrounding domain through copper finned heat sink with fin spacing, fin height, baseplate thickness and fin thickness as 0.05 m, 0.03 m, 0.02 and 0.02 m respectively. Figs. 9 and 10.

Fig. 11 depicts the temperature contours along the axial plane of PV panel with maximum temperature of 68.78 °C at the panel surface and minimum temperature of 42 °C inside the ventilated channel. The average operating temperature of 50 °C was obtained.

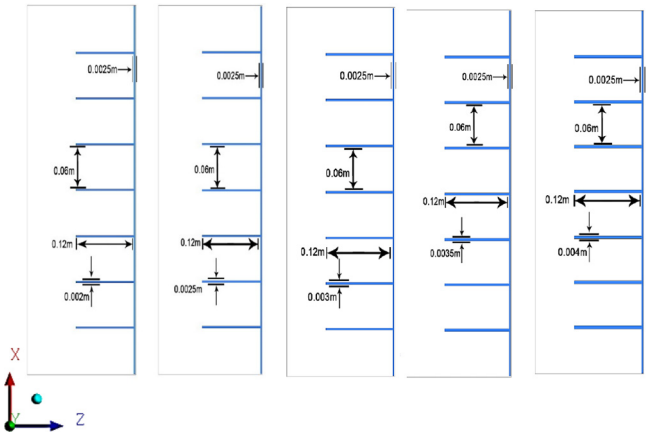


Fig. 10. Geometry of simulated heat sink integrated PV models with varying fin thickness a)0.002 m, b)0.0025 m,c) 0.003 m, d)0.0035 m and e)0.004 *m.*

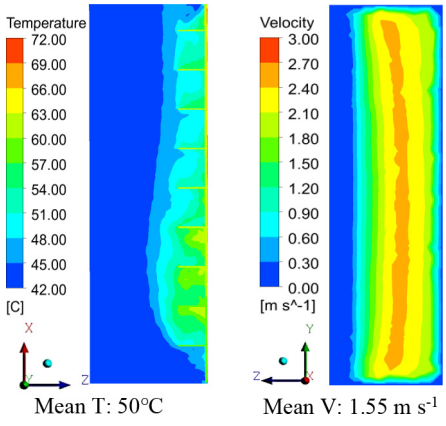


Fig. 11. Temperature (left) and velocity (right) contours of passively cooled PV panel using base model heat sink.

The velocity contours at the outlet plane show high velocity/turbulent airflow of 2.55 m/s, near the fin edges.

4.2. Parametric study's findings

4.2.1. Case 1 – Varying fin spacing

The results for models with varying fin spacings are shown in Figs. 12 and 13. The maximum temperatures on the external PV cell layer are 69 °C, 69.69 °C, 68.78 °C, 71 °C, 69.12 °C and 68 °C for fin spacings 0.03 m, 0.04 m, 0.05 m, 0.06 m, 0.07 m, and 0.08 m, respectively, at minimum temperature of 42 °C (within ventilated channel). From Fig. 13, the average outlet velocities of the models increase from 1.56 m/s to 1.52 m/s. The maximum velocities at the highest thermal transfer region were 2.92 m/s, 2.60 m/s, 2.55 m/s, 2.46 m/s, 2.43 m/s and 2.37 m/s respectively.

4.2.2. Case 2 – Varying baseplate thickness

The CFD-Post processor conotours with maximum temperatures at 68.86 °C, 69.47 °C, 69.30 °C, 69.21 °C and 69 °C for the respective base thicknesses are shown in Fig. 14. The velocity profiles at outlet location depicted a gradual increase in average velocities from 1.54 m/s to 1.55 m/s. The maximum velocities at the highest thermal transfer region were 2.46 m/s, 2.54 m/s, 2.51 m/s, 2.47 m/s and 2.46 m/s respectively.Fig. 15.

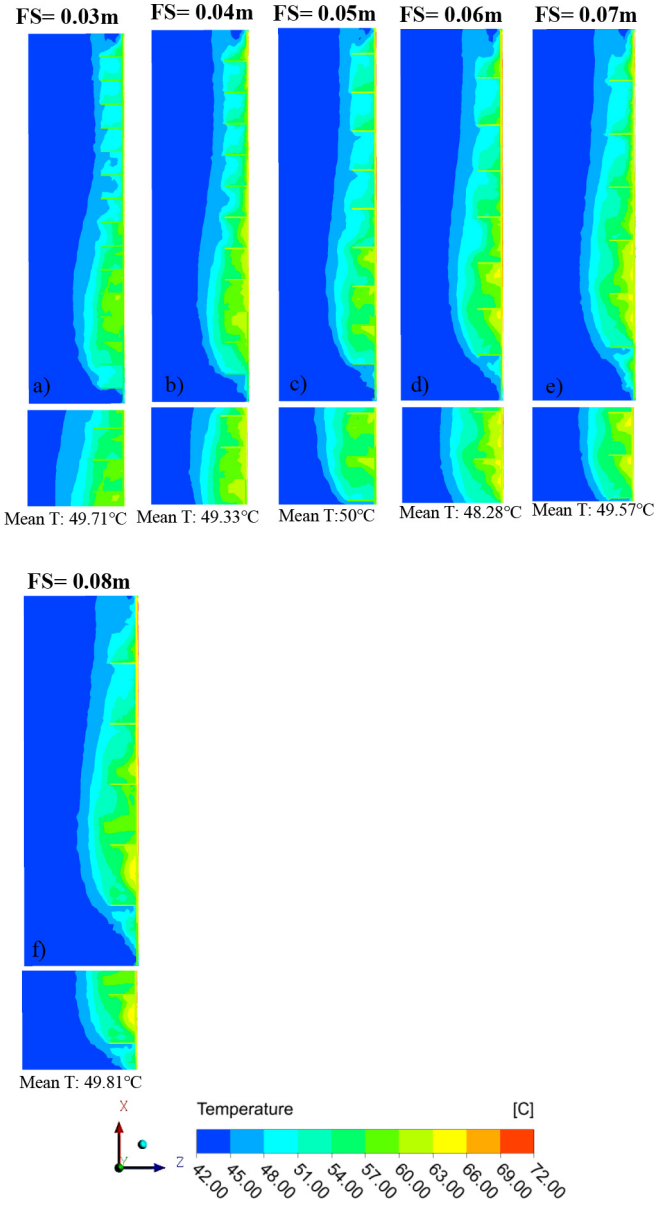


Fig. 12. Temperature contours of passive cooled PV panel using copper heat sinks with varying fin spacing as a)0.03 m, b)0.04 m, c) 0.05 m, d)0.06 m e)0.07 m and f) 0.08 *m.*

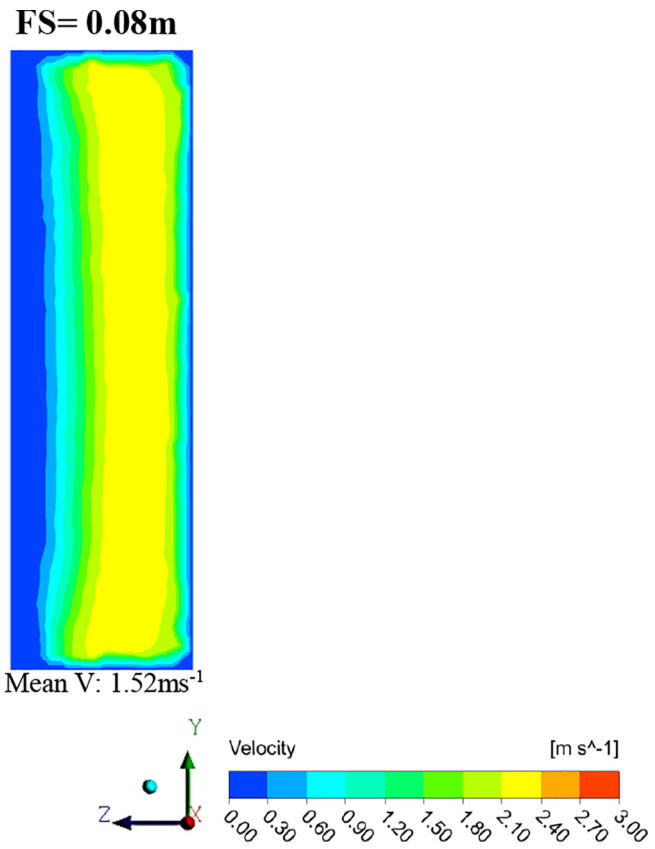


Fig. 13. Outlet Velocity profiles for varying fin spacing a)0.03 m, b)0.04 m, c) 0.05 m, d)0.06 m e)0.07 m and f)0.08 *m.*

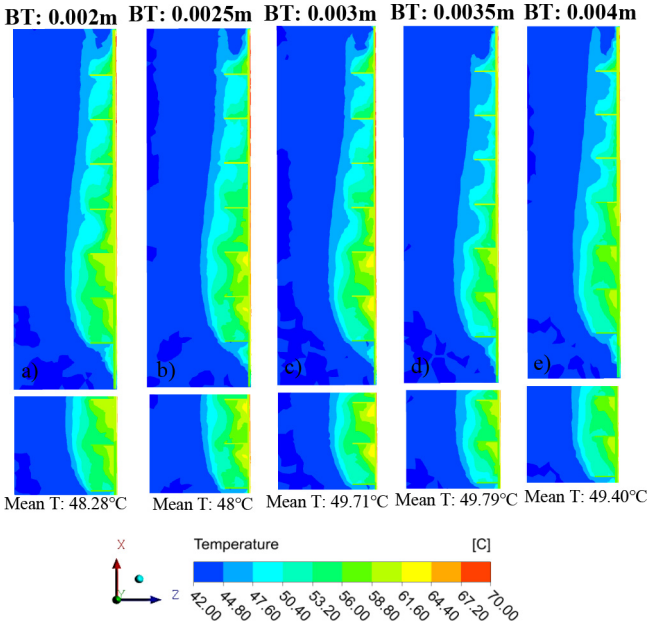


Fig. 14. Temperature contours of passive cooled PV panel using copper heat sinks with varying base thickness as a)0.002 m, b)0.0025 m, c) 0.003 m, d)0.0035 m and e) 0.004 *m.*

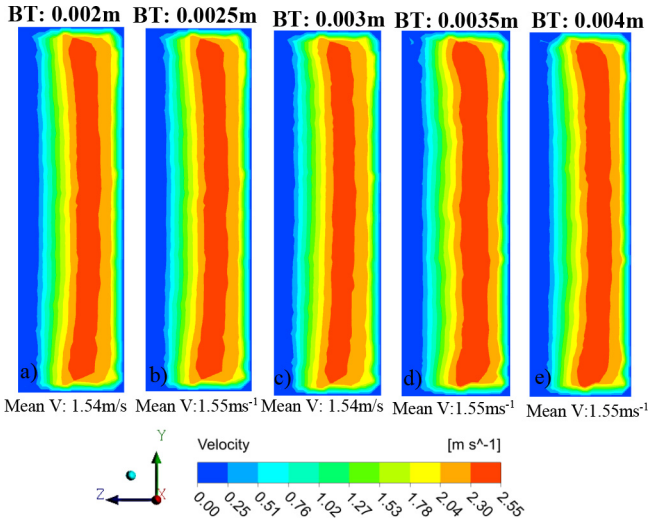


Fig. 15. Outlet Velocity profiles for varying base thickness a)0.002 m, b)0.0025 m, c) 0.003 m, d)0.0035 m and e)0.004 *m*.

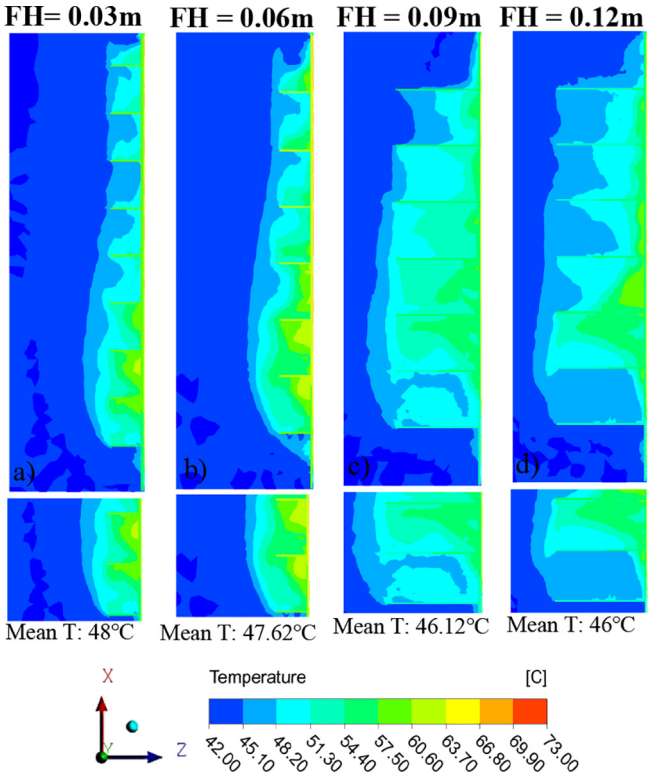


Fig. 16. Temperature contours of passive cooled PV panel using copper heat sinks with varying fin heights as a)0.03 m, b)0.06 m, c) 0.09 m and d)0.12 *m*.

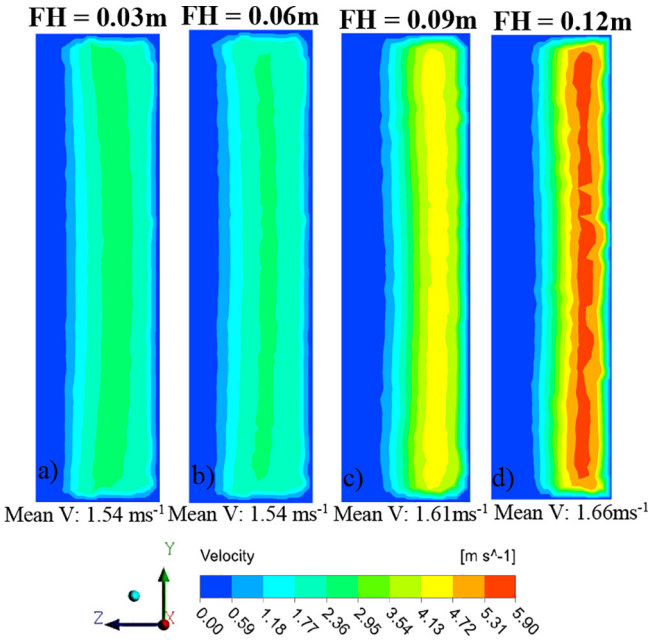


Fig. 17. Outlet Velocity profiles for varying fin heights a)0.03 m, b)0.06 m, c) 0.09 m and d)0.12 *m*.

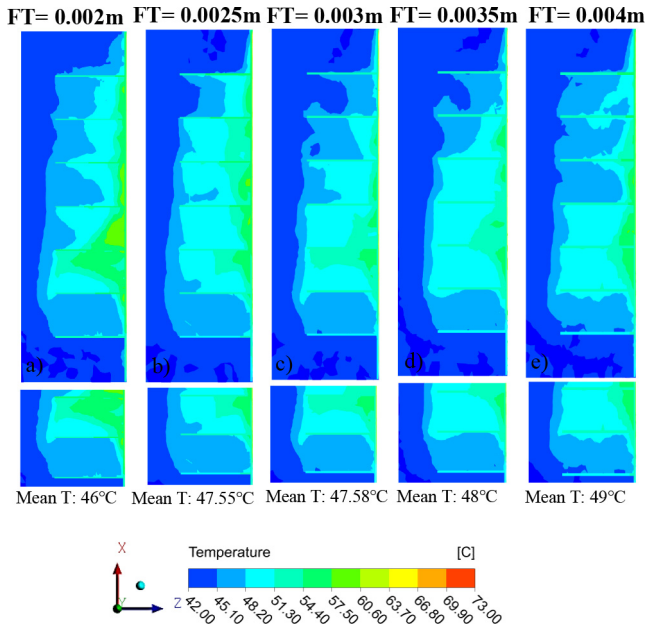


Fig. 18. Temperature contours of passive cooled PV panel using copper heat sinks with varying fin thickness as a)0.002 m, b)0.0025 m, c) 0.003 m, d)0.0035 m and e) 0.004 *m*.

4.2.3. Case 3 – Varying fin height

The simulation results for models with increasing fin heights are shown below. From the temperature contours, maximum surface temperatures of 69.47 °C, 69.82 °C, 70 °C and 72 °C were obtained for increasing fin heights. The outlet velocities increased with increase in fin height. From the contours in Fig. 17, the maximum outlet velocities are 2.47 m/s, 2.49 m/s, 4.52 m/s and 5.70 m/s for the respective fin heights.Fig. 18.

4.2.4. Case 4 – Varying fin thickness

The maximum temperatures at external surface of the models are 72 °C, 68 °C, 65.50 °C, 65 °C and 63.90 °C for fin thickness 0.002 m, 0.0025 m, 0.003 m, 0.0035 m, and 0.004 m, respectively. The average outlet velocities gradually reduce from 1.66 m/s to 1.08 m/s. Similarly, maximum velocities at the highest thermal transfer region were 5.70 m/s, 5.68 m/s, 5.64 m/s, 5.58 m/s and 5.52 m/s respectively.

4.2.5. Case 5 – Material properties

The following temperature and velocity contours at the axial and outlet plane were obtained for the aluminium heat sink model having 0.06 m fin spacing, 0.0025 m baseplate thickness, 0.12 m fin height, 0.002 m fin thickness. The temperature range across the model geometry goes from 61 °C at PV panel surface to 42 °C within the airflow channel. Meanwhile, the total average temperature lies at 47 °C. The velocity contours depict a high velocity of 5.70 m/s at the center with average velocity of 1.25 m/s.

5. Discussion

5.1. Cooling performance of base model heat sink

The simulation results of using a copper heat sink with fin spacing, fin height, baseplate thickness and fin thickness as 0.05 m, 0.03 m, 0.002 m and 0.002 m depicted that the high surface temperature of the PV panel can be successfully reduced by about 27 °C, from 68.78 °C to 42 °C. The heat transfer through fins was aslo evident from a high turbulence area (V_{max} = 2.55 m/s) close to the fin edges, depicting a region of maximum thermal transfer between the two media (air and copper) as the heat transfer coefficient increases with increase in airflow velocity. Further investigation of internal temperatures across axial and tangential directions through the fin geometry ascertained the exponential decrease in temperatures due to accelerated heat transfer. From the line graphs below (Fig. 22), maximum temperature drop of 17 °C and 27 °C were observed along X and Z axes.

5.2. Cooling performance comparison for varying physical components

The effect of varying fin spacing on the heat dissipation rate of copper heat sinks is such that with fin spacing increasing from 0.03 m to 0.06 m, the heat dissipation rate increased gradually but reduced when spacing increased beyond 0.06 m. The 7 finned arrangement spaced 0.06 m apart has the highest temperature reduction (ΔT), from 42 °C to 71 °C. An improvement of 1.22 °C in the cooling performance was seen when compared to base case scenario with 0.05 m of fin spacing (9 fins). The same tendency of optimum cooling was depicted by comparing the overall PV panel temperatures of optimum model with base model showing a drop of 1.41 °C from 50 °C to 48.28 °C. The simulation results therefore indicated that when the surface of the heat sink does not sufficiently satisfy the requirements/criteria for heat dissipation, it will worsen the cooling performance. A wider gap between the fins reduces the area for heat transfer but, an extremely smaller gap will in turn lead to insufficient space for the air convection to occur through the fins. The high turbulence velocity contours for smaller fin spacings, corresponds to frequent variations in airflow speed and direction causing high turbulence from increased friction between fins. Further analysis of internal temperatures across the axial and tangential directions (Figs. 24 and 25) also proved a higher and much uniform rate of heat dissipation for the model with 7 fins since the lowest internal temperatures of 49 °C and 44 °C were obtained.

The heat dissipation rate of PV panels changes only slightly with increasing base thickness, the difference between highest and lowest temperature drop was only 0.6 °C. Fig. 23 for varying baseplate thickness shows that the heat sink with 0.0025 m thick copper base provided a maximum temperature drop of 27.47 °C. The graph shows that a thicker plate can clearly improve heat dissipation between the panel and surrounding medium but a thickness beyond 0.0025 m gave slightly diminishing results due to the fact that heat transfer by conduction depends on close bonding between transfer-interface and heat source. An extremely large thickness will in turn impede the heat transfer also causing material wastage and additional space requirements. Therefore, the 0.0025 m thickness was imperative in obtaining optimum cooling performance and was used for the subsequent parametric study of varying fin heights. Temperature ranges across the axial and tangential axes of the five models also proved maximum temperature reduction for the 0.0025 m thick baseplate model in addition to a much uniform temperature range across × and z axes as evident by lower peaks in the line graphs (Figs. 24 and 25). The average internal temperatures along X and Z axes also exponentially dropped by 6 – 1 °C and 1.5 – 0.6 °C for the optimum heat sink model.

For the subsequent parametric study, the fin height was increased by 1% for each model depicting an accelerated thermal dissipation rate. The cooling performance increases with increase in fin height due to increase in the cross-sectional area for heat rejection. The maximum temperature reduction across the panel geometry was 30 °C for the fin height of 0.12 m. A difference of around 3 °C temperature drop was observed for 3% height increment from 0.03 m to 0.12 m. The fin height parameter is highly significant in optimizing the heat sink performance as the high surface temperatures of the optimum model dropped by 3 °C compared that of base model heat sink. A drop of 0.5 °C-1.5 °C in the overall PV panel temperature was obtained as the temperatures reduced from 48 °C, 47.62 °C, 46.12 °C to 46 °C for respective fin heights in Fig. 16. However extremely high fins will require larger area thereby impacting air convection within the congested space (between rear surface and mounting base). Additionally, higher fins will also be fragile and easy to fracture from greater standing stress of the material and high-speed winds. The corresponding velocity contours in Fig. 17 determined a rapid increase in the mean airflow velocity and higher turbulence bands (V_{max} = 5.70-m/s for 0.12 m) close to fin edges. Delving into the internal temperatures across axial and tangential directions of the models also validated the exponential increase in heat dissipation in higher fins. The line graphs in Figs. 24 and 25 depict uniform temperature reductions for 0.12 m fin height compared to other models that reduces thermal stress and heat spots on the cell surface thus improving the overall panel efficiencies and lifespan.

Lastly, the most optimum heat sink structure was obtained by increasing the fin thickness by 0.25% for each model, observing lower temperature drop for increasing fin thicknesses. The most optimum fin thickness of 0.002 m was concluded as the panel temperature dropped by 30 °C. Overall operating temperatures of the five models progressively increased from 46 °C to 49 °C as the thickness increased from 0.002 m to 0.004 m. This is because an increasing fin thickness reduces the intervening fin spacing that in turn reducing the space for sufficient air convection around the fins. Further, the conductive heat transfer also reduces with increasing thickness as the heat flow from panel surface to surrounding medium will be over longer distance causing insufficient heat transfer. This optimum heat sink structure therefore reduced the total PV panel temperature by 4 °C compared to the base model of copper heat sink. The velocity contours in Fig. 19 determine a rapid decline in the mean airflow velocity from 1.66 m/s to 1.08 m/s, also signifying lower heat transfer. The average internal temperatures of models with thicker fins reduced by 3 °C and 1 °C along axial and tangential directions.Figs. 20 and 21.

5.3. Cooling performance comparison for varying elemental properties

Since optimization is the key to an efficient heat sink, the elemental properties of the most optimum copper heat sink structure were further investigated using CFD analysis by altering its material to aluminium. Aluminium heat sinks are commonly used due to their cost and weight saving properties but have a lower thermal

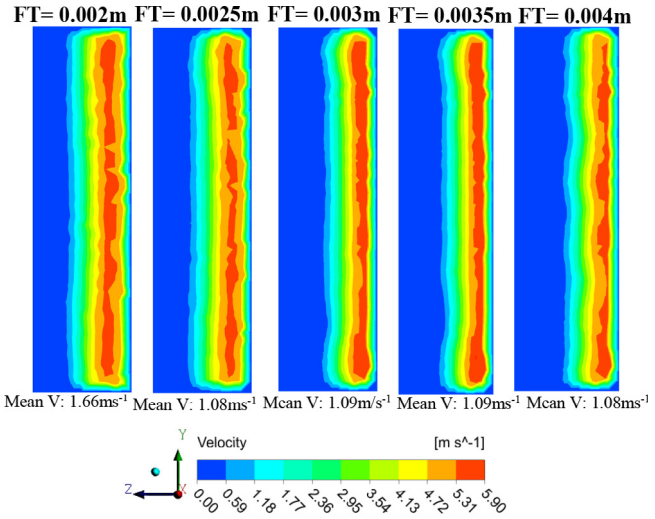


Fig. 19. Outlet Velocity profiles for varying fin thickness as a)0.002 m, b)0.0025 m, c) 0.003 m, d)0.0035 m and e) 0.004 *m*.

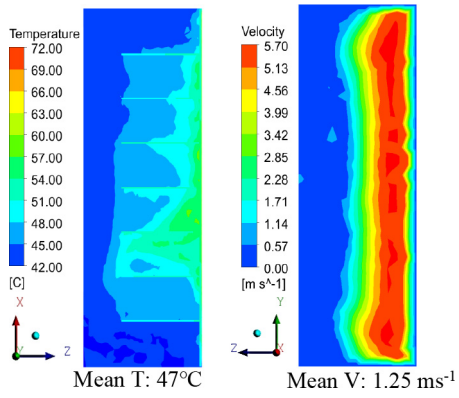


Fig. 20. Temperature and velocity contours of passive cooled PV panel using aluminium heat sink with optimized heat sink model.

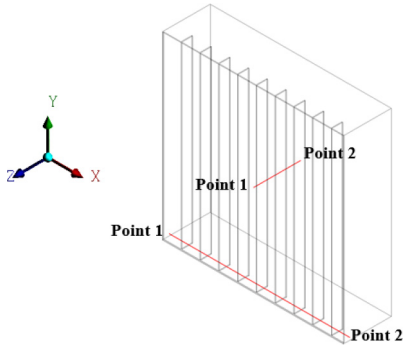


Fig. 21. Axial and tangential lengths position on the geometry of simulated model.

conductivity compared to copper, which is responsible for inefficient and slower transport of heat energy through the material lattice therefore leading to poorer heat dissipation from the PV panel surface to surrounding medium [24]. As seen from the above graph, the copper heat sink could reduce the surface temperatures by 30 °C compared to aluminium heat sink that only reduced 19 °C.

The results depicted that the overall temperature of copper heat sink integrated PV panel was 1 °C lower than the aluminium heat sink studied under the same ambient conditions therefore, concluding copper heat sinks to be advantageous over aluminium heat sinks. On comparing the internal temperatures of both models as shown below, the maximum temperature drop by copper heat sink along the axial and tangential lengths was found to be 11 °C higher than that of aluminium. Although, both the models show a uniform temperature reduction through the use of optimized heat sink configuration, the copper heat sink can dissipate 2.3% and 2.4% more heat along the two directions compared to the aluminium heat sink.

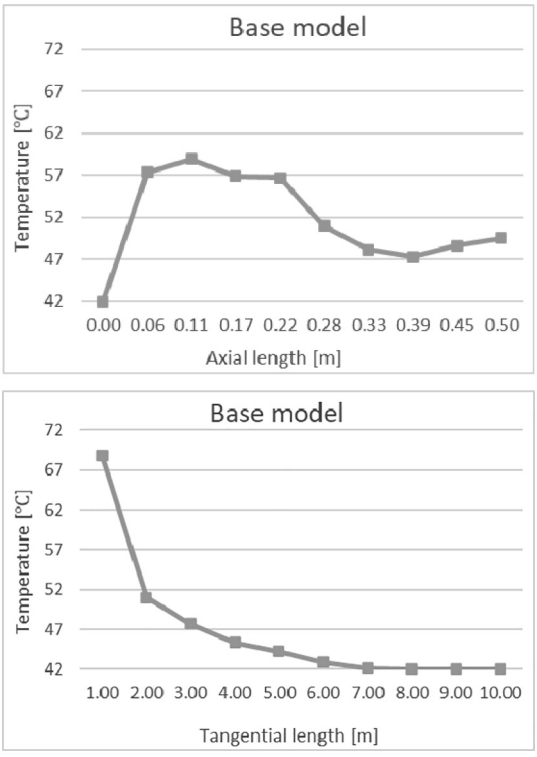


Fig. 22. Internal temperature variations along the axial (left) and tangential (right) lengths.

The dynamic simulations from ANSYS CFD software enabled us to point out the regions on heat sinks with insufficient fluid flow and poor heat dissipation. It provided a visual and numerical description of the flow field and temperature distribution in and around the heat sink inside the enclosure. The information about the fluid velocities near the wall in thermal boundary layer enhanced the accuracy of interpreting the heat transfer from the heat sink to the surrounding domain. The combination of temperature distribution and airflow distribution helped to decide the necessary design changes to achieve the optimium heat sink performance through an iterative proces.Figs. 26 and 27.

5.4. Electrical performance enhancement of passively cooled PV panel

The Fig. 28 shows reducing operating temperatures of the PV panel for optimized geometries. As each structural component was optimized the average temperatures and surface temperatures dropped due to increase in the heat dissipation properties of the copper heat sink. The overall temperature drop for the most advantageous heat sink model is 9% compared to the base model of heat sink. Additionally, the maximum temperature drop across the simulated PV system body is 12% higher than the base case scenario. Moreover, it is observed that the baseplate thickness parameter is not sufficiently significant in optimizing the cooling performance as above trend of surface temperature reduction shows a minor

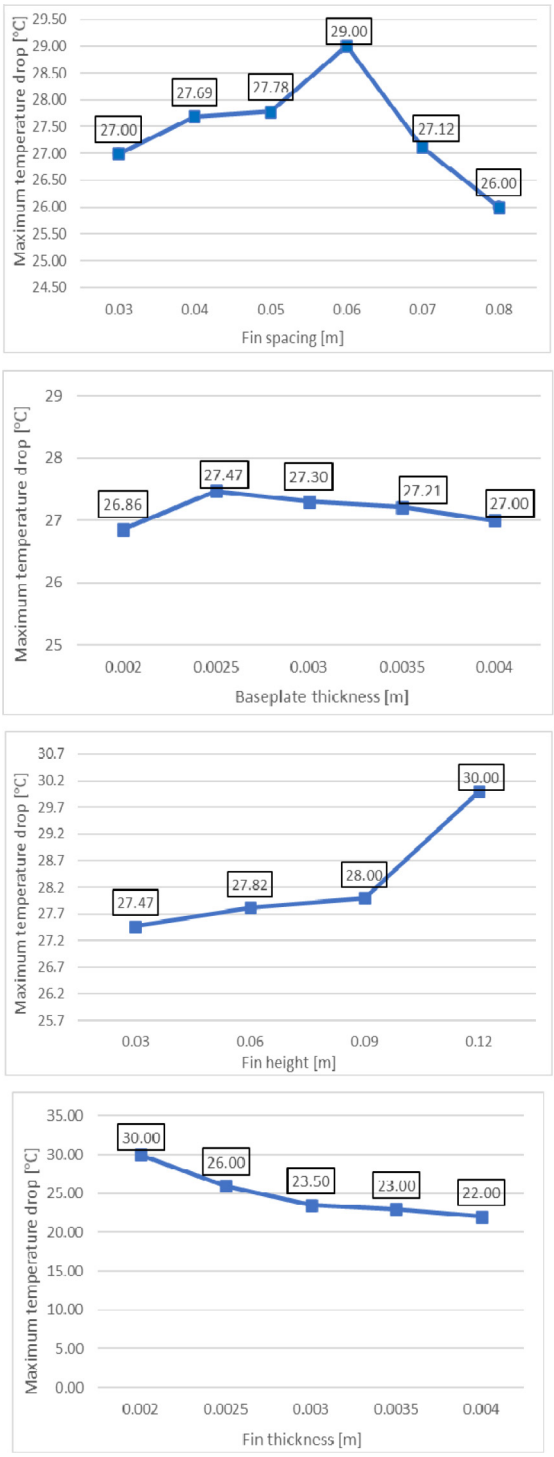


Fig. 23. Cooling performance summary of heat sink models with varying fin spacing, baseplate thickness, fin height and fin thickness.

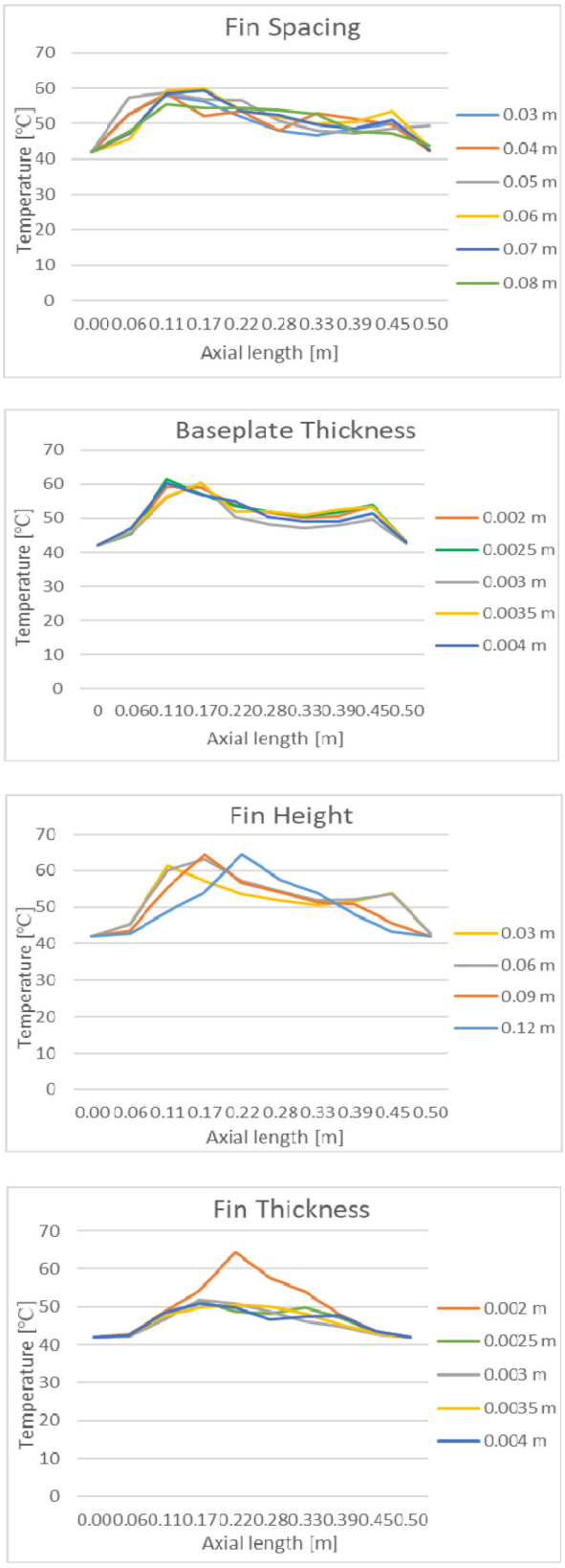
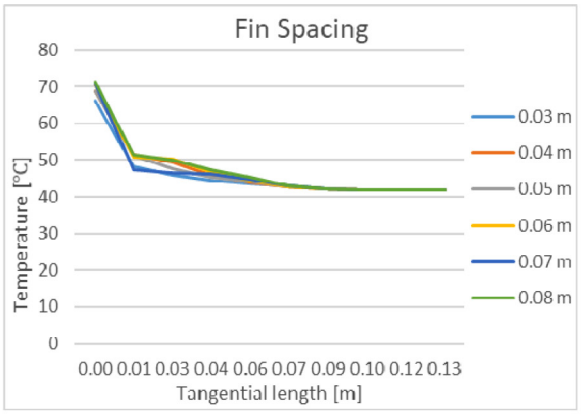
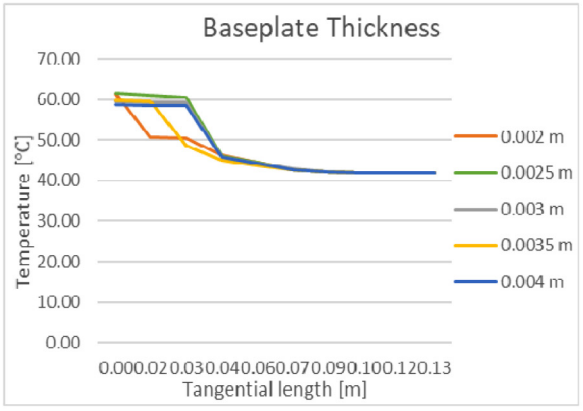


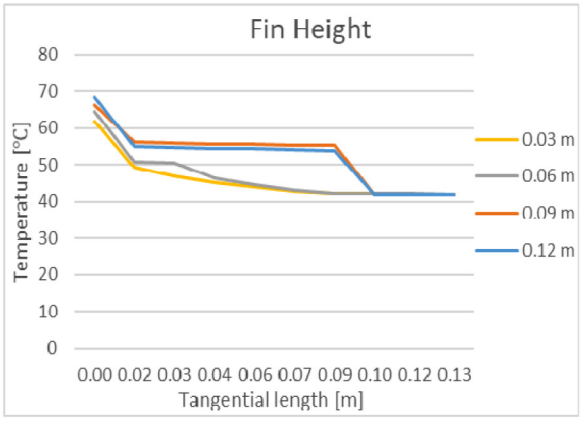
Fig. 24. Internal temperature variations along the axial length for models with varying fin spacings, baseplate thickness, fin height and fin thickness.

drop compared to other parameters. The fin arrangement and fin height however play a major role in accelerating the dissipation rate. Further, the analysis of heat sink material properties depicted copper to be more advantageous than the commonly used aluminium that increased the panel temperature by 2%, despite the integration of optimized model.

This benefit of passive cooling on the electrical performance of a polycrystalline PV panel is represented by the following Table 8 and Fig. 29. Maximum increase in electrical efficiency and power output of the panel by 0.18% can be achieved by structural and material optimization. The reduction in STC panel efficiency after structural optimization is only 0.61% compared to 0.80% with the







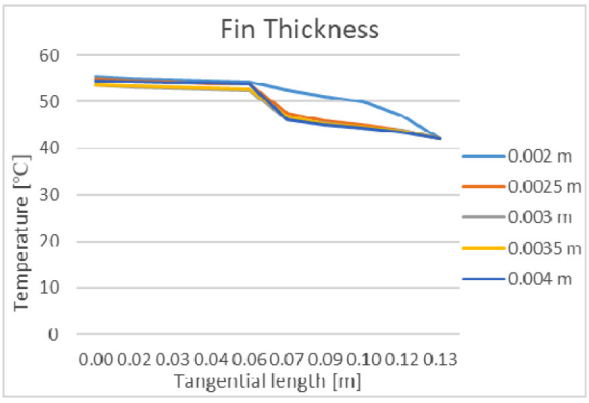


Fig. 25. Internal temperature variations along the tangential length for models with varying fin spacings, baseplate thickness, fin height and fin thickness.

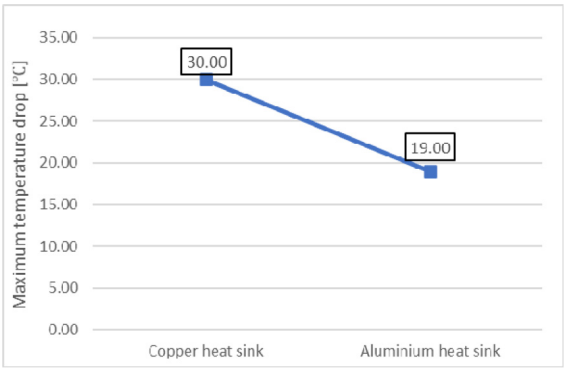
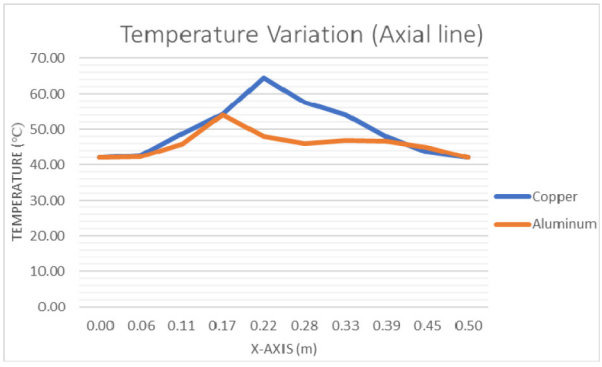


Fig. 26. Cooling Performance Summary of optimized copper and aluminium heat sinks.



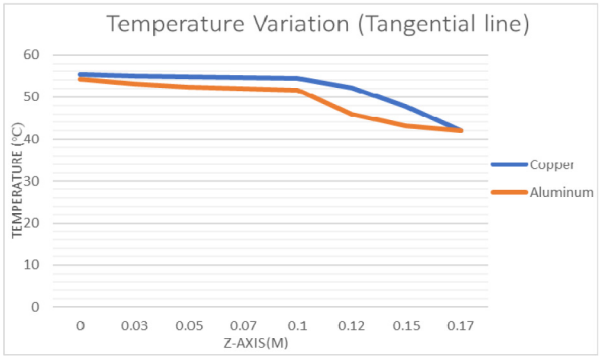


Fig. 27. Internal temperature variations along the axial (top) and tangential (bottom) length for copper and aluminium heat sinks.

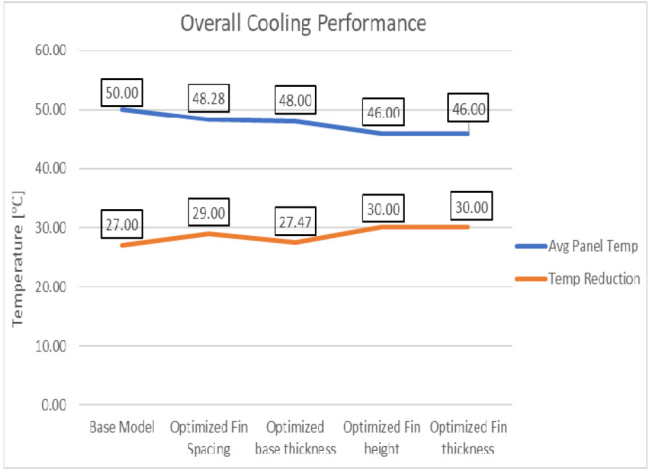


Fig. 28. Overall Cooling Performance comparison after optimization of copper heat sink.

Table 8
Influence of passive cooling over efficiency and power enhancement of Polycrystalline PV panel.

Model	Average Panel Temperature (°C)	Efficiency (%)	Power Output (W)
Base Model	50	16.07	3616
Optimized Fin Spacing	48.28	16.09	3620
Optimized Baseplate Thickness	48	16.09	3618
Optimized Fin Height	46	16.10	3622
Optimized Fin Thickness	46	16.10	3622

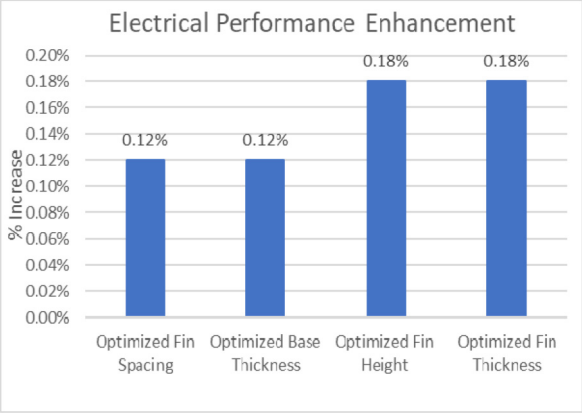


Fig. 29. Influence of passive cooling over electrical performance of Polycrystalline PV panel.

base model. The conclusion of a higher thermally conductive metal, copper was confirmed as studying the aluminium heat sink model further dropped the efficiency by 0.74%.

6. Conclusion

In order to encourage implementation of building-integrated solar panels, the PV cells need to operate close to a nominal operative temperature as much as possible. This is a particular challenge in countries facing very hot and humid climates all year around. This work carries out an extensive numerical modelling investigation to understand the implications of having an air-cooled sink retrofitted to photovoltaic panels to depict the improvements in the working performance of the PV cell. The conclusions of the study are listed below.

- With the increasing fin spacing, the temperature of solar cells first decrease and then increase stating an optimal spacing of 0.06 m with 7 fins. This fin arrangement produced the highest temperature drop of 29 °C and lowest mean panel temperature of 48.28 °C. The first parametric study resulted in increase of overall temperature drop by 3.5% compared to base heat sink structure.
- Similarly, increasing the baseplate thickness also reduced the panel temperature but gave diminishing results for a thicker base. The thickness of 0.0025 m was concluded optimum as it depicted a higher temperature drop of 27.47 °C compared to thicker plates. The optimum thickness depicted about 4% increased reduction in the average panel temperature at 48 °C.
- Temperature of the PV panel exponentially decreased with increase in fin height and 0.12 m high fins represented the highest drop of 30 °C thereby, massively reducing the panel temperature to 46 °C. The cooling performance of model with high fins increased by 9% compared to base case scenario of 0.03 m fin height.
- Increasing the fin thickness in turn increases panel temperatures, therefore the most optimum heat sink model with default fin thickness of 0.002 m was obtained with the average panel temperature constant at 46 °C.
- Altering the material of copper heat sink to aluminium worsened the heat dissipation rate as the average panel temperature increased by 2% at 47 °C, despite the use of optimal heat sink geometry.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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